

An approximation-property subcase of the weak-null tensor question

FA/Banach run packet for arXiv:2305.06089

2026-06-28

Source question

The source paper is J. Rodriguez and A. Rueda Zoca, *Weak precompactness in projective tensor products*, arXiv:2305.06089. In Question **Gonzalo**, the authors ask:

Let X and Y be Banach spaces. Let (x_n) and (y_n) be weakly null sequences in X and Y , respectively, such that $(x_n \otimes y_n)$ is weakly convergent in $X \widehat{\otimes}_\pi Y$. Is $(x_n \otimes y_n)$ weakly null in $X \widehat{\otimes}_\pi Y$?

Idea of the proof

If $x_n \rightarrow 0$ weakly and $y_n \rightarrow 0$ weakly, then every elementary tensor functional sees $x_n \otimes y_n$ going to zero:

$$(x^* \otimes y^*)(x_n \otimes y_n) = x^*(x_n)y^*(y_n) \rightarrow 0.$$

Thus any weak limit of the tensors is invisible to all elementary tensor functionals. The only possible hiding place is the kernel of the canonical map from the projective tensor product to the injective tensor product. When this kernel is zero, for instance when one tensor factor has the approximation property, the weak limit must be zero.

The theorem

Theorem 1. *Let X and Y be Banach spaces, and let*

$$J : X \widehat{\otimes}_\pi Y \longrightarrow X \widehat{\otimes}_\varepsilon Y$$

be the canonical contractive map. Suppose that J is injective. If $(x_n) \subset X$ and $(y_n) \subset Y$ are weakly null and $(x_n \otimes y_n)$ is weakly convergent in $X \widehat{\otimes}_\pi Y$, then $(x_n \otimes y_n)$ is weakly null.

*Consequently, Question **Gonzalo** has an affirmative answer whenever either X or Y has the approximation property.*

Proof. Let $u \in X \widehat{\otimes}_\pi Y$ be the weak limit of $x_n \otimes y_n$. Fix $x^* \in X^*$ and $y^* \in Y^*$. The functional $x^* \otimes y^*$ is continuous on $X \widehat{\otimes}_\pi Y$, and hence

$$(x^* \otimes y^*)(u) = \lim_n (x^* \otimes y^*)(x_n \otimes y_n) = \lim_n x^*(x_n)y^*(y_n) = 0,$$

because both scalar sequences $x^*(x_n)$ and $y^*(y_n)$ converge to zero.

The injective norm is computed by elementary dual tensors:

$$\|Jv\|_\varepsilon = \sup \{ |(x^* \otimes y^*)(v)| : x^* \in B_{X^*}, y^* \in B_{Y^*} \}.$$

Applying this to $v = u$, the previous paragraph gives $\|Ju\|_\varepsilon = 0$. Hence $Ju = 0$. Since J is injective, $u = 0$. Therefore $x_n \otimes y_n$ converges weakly to zero.

For the final assertion, use the standard tensor-norm characterization of the approximation property: if either factor has the approximation property, then the canonical map $X \widehat{\otimes}_\pi Y \rightarrow X \widehat{\otimes}_\varepsilon Y$ is injective for every choice of the other factor. $\square$

Remark 1. *The proof actually identifies the only possible obstruction in the general case. If Question Gonzalo has a negative answer, then the weak limit of a counterexample must be a nonzero element of $\ker J$. Thus non-injectivity of the projective-to-injective map is necessary for any counterexample, though not by itself sufficient.*

Verification and limitations

The argument uses only weak convergence against elementary tensor functionals and the standard approximation-property injectivity criterion for the projective-to-injective canonical map. It does not address pairs (X, Y) for which J has nonzero kernel; that is the remaining hard region of the source question.

Novelty check

The cheap run indexes were searched for arXiv:2305.06089, the source title, weak precompactness, projective tensor products, weakly null tensor sequences, and related Dunford–Pettis/projective-tensor keywords. Web searches for the source title, exact question phrases, and “weakly null $x_n \otimes y_n$ projective tensor approximation property” did not surface an exact later answer. This packet should be reviewed as a modest but clean partial result.

References

- J. Rodriguez and A. Rueda Zoca, *Weak precompactness in projective tensor products*. arXiv:2305.06089.
- R. A. Ryan, *Introduction to Tensor Products of Banach Spaces*, Springer, 2002. Used for standard projective/injective tensor-product facts and the approximation-property injectivity criterion.

Human-review recommendation

Review as a likely valid partial result. The main points to check are the standard AP criterion for $X \widehat{\otimes}_\pi Y \rightarrow X \widehat{\otimes}_\varepsilon Y$ and that elementary tensor functionals determine the injective norm exactly as used above.